A’di Dust

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Los Angeles, CA

Ph.D. Student
University of Southern California
Department of Computer Science

EDUCATION

***Ph.D.*, Computer Science, University of Southern California**
June 2023 - present. Advisor: Prof Maja Matarić. Collaborating Advisor: Prof Pat Levitt.

***B.A.*, Computer Science, Honors and Distinction, Macalester College**
August 2019 - May 2023. Konhauser Achievement Award in Computer Science. Honors thesis: *Using the Social Model of Disability Studies as a Framework to Assess Deaf-Centered Technology*. Advisors: Prof Abby Marsh and Prof Joslenne Peña. Minors in Sociology and Statistics.

PUBLICATIONS AND PRESENTATIONS

A’di Dust, Anura Deshpande, Maja J. Matarić, and Pat Levitt. “Signals in Play: Maternal Emotional State Modeled Using Conversational Dynamics and Affective Prosody.” *Under review.*

A’di Dust, Sarah Panvini, Maja J. Matarić, and Pat Levitt. “Advanced Analysis of Behavioral Synchrony and Parasympathetic Response in Mother-Infant Play.” *Oral presentation at the 2025 Annual Meeting of the International Society of Developmental Psychobiology,* San Diego, CA, USA.

Manal Tabbaa, Alexis Gamez, **A’di Dust**, Maja Matarić, Pat Levitt. “Offspring Genetic Diversity Regulates Rearing Experiences that Predict Differential Susceptibility to Chd8 Haploinsufficiency.” *In press at Communications Biology.*

A’di Dust, Pat Levitt, and Maja J. Matarić. “Behind the Smile: Mental Health Implications of Mother-Infant Interactions Revealed Through Smile Analysis.” *In the 12th International Conference on Affective Computing and Intelligent Interaction (ACII ‘24)*. Glasgow, SCT, September 2024.

A’di Dust, Carola Gonzalez-Lebron, Shannon Connell, Saurav Singh, Reynold Bailey, Cecilia Ovesdotter Alm, and Jamison Heard. “Understanding Differences in Human-Robot Teaming Dynamics between Deaf/Hard of Hearing and Hearing Individuals.” *In Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (Companion-HRI '23)*, Stockholm, SE, March 2023.

A’di Dust. “Hearing in Cobot Environments: The Effects on Deaf Workers in a Manufacturing Setting.” *RIT Undergraduate Research Symposium*. Poster Presentation, Rochester, NY, July 2022.

A’di Dust. “Measured Alteration Of Generalized Procedural Generation For Robot Agility.” *NIST Summer Undergraduate Research Fellowship Colloquium.* Oral Presentation, Virtual, August 2021.

RESEARCH EXPERIENCE

University of Southern California Interaction Lab

June 2023 - present. Ph.D. Student, Advisor*:* Prof Maja Matarić.

1. Conducting interdisciplinary research with Children’s Hospital Los Angeles (Levitt Lab) to analyze mother-infant behavioral synchrony and its associations with developmental and physiological health outcomes.
2. Applying computational modeling, signal analysis, and human-robot interaction methods to study early-life behavioral markers.
3. Developing a social emotional learning robotic platform for early childhood support, integrating sensing, behavior modeling, and adaptive interaction algorithms.
4. Contributed to an NSF Convergence Accelerator project focused on enhancing social support for individuals with physical disabilities through advanced assistive and interactive technologies.

Rochester Institute of Technology Adaptive Human-Robot Teaming Lab

May 2022 - July 2022. Undergraduate Researcher. Advisor: Prof Jamison Heard.

1. Selected to research in the NSF-funded Computational Sensing for Human-centered AI program.
2. Conducted human subjects research on Deaf and Hard-of-Hearing individuals in relation to their interactions with manufacturing cobots to understand differences in cobot teaming compared to hearing individuals.
3. Programmed the Baxter cobot and performed detailed data analysis of human subject trials.
4. Used a variety of computational sensing tools such as Pupil Core eyeglasses and Zephyr bioharness.

National Institute of Standards and Technology Intelligent Systems Division
May 2021 - August 2021. Associate Research Fellow. Advisor: Dr. William Harrison III.

1. Selected to conduct research for the NIST Summer Undergraduate Research Fellowship's (SURF) Engineering Laboratory.
2. Researched generalizing procedural generation of 3D environments in which to simulate reinforcement learning for robot safety.
3. Created an original method called Measured Alteration of Generalized Procedural Generation for Robot Agility
4. Created a Unity virtual prototype to simulate the alteration in a realistic manufacturing environment.

TEACHING AND INDUSTRY EXPERIENCE

Macalester College
September 2020 - May 2023. Teaching Assistant.

1. Fall 2020. COMP 127: Object Oriented Programming and Abstraction.
2. Spring 2021, Fall 2021, Spring 2022, Fall 2022, Spring 2023. COMP 240: Computer Systems Organization.

Civilytics Consulting LLC
July 2020 - June 2021. Contractor.

1. Created content and managed social media for data science in higher education, K-12 education, criminal justice, and public finance projects
2. Collected and cleaned data from a variety of sources including city government budgeting documents and campus policing documents
3. Formatted Education Data Done Right II book using R Bookdown

MENTORING

Master's Students: Praise Olukilede (2023), Zoey Zhou (2023), Jiahang Zhang (2024).

Undergraduate Students: Anura Deshpande (2025 -), Cindy Zhou (2025 -), Evelyn Gonzalez (2025 -), Soumya Singh (2025), Kyla Penamante (2024 -), Parini Gandhi (2024 -), Elisa Torres (2024), Amanda Lu (2023 - 2025), Dylan Gatua (2023 - 2025), Vincent Wong (2023).

K-12 Students: Nina Smith (2025-), Naman Rathi (2025), Natalia Colin (2024), Joan Delgadillo (2024).

SERVICE

Kory Hunter Middle School Vex Robotics
November 2023 - March 2024. Coach.

1. Taught fundamentals of designing hardware and software for Vex Robotics competition.

Flipside Coding Club
January 2023 - May 2023. Teacher.

1. Taught Python, HTML, JavaScript, and CSS to students from 4th to 7th grade.
2. Worked on building skills related to thinking about disabilities and accessibility when creating software.

Breakthrough Twin Cities
October 2020 - May 2021. Coach.

1. Mentored 15 disadvantaged 7th grade students through the school year.
2. Set up monthly meetings with each student, along with weekly email check-ins regarding home life, school, and goals.
3. Facilitated teaching lessons about career possibilities and best practices.

Twin Cities Reading Partners
October 2019 - March 2020. Instructor.

1. Mentored one low-literacy middle school student.
2. Created custom lesson plans each week and facilitated learning reading and writing skills in each meeting with the student.

Kidz Klub
September 2017 - May 2019. Afterschool Caregiver.

1. Worked with elementary students from disadvantaged backgrounds after school.
2. Created activities and facilitated homework with children each day.
3. Occasionally hosted field trips and educational experiences during school breaks.

Spring Hill Elementary
September 2016 - May 2019. Assistant Music Teacher.

1. Assisted kindergarten through second grade music teachers with each music class.

SKILLS AND CERTIFICATES

Languages and Frameworks: Python, R, Java, C, C#, JavaScript, Swift, LaTeX, scikit-learn, pandas, NumPy, PyTorch, OpenCV, DEAP, SPSS, ggplot2, Altair, D3.js, ROS, Linux, robot learning frameworks, generative learning models

Database Management: Firebase and Firestore

Certifications: CITI Human Subjects Research, CITI Responsible Conduct of Research, HIPAA Compliance

Other Skills: Machine learning, Data visualization, Physiological and behavioral data acquisition, Human-subjects research design, Affective computing, Robot learning